

PHAM DUC LONG

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SUMMARY

Final-year Software Engineering student focused on backend development, with hands-on experience building Spring Boot applications and RESTful APIs. Familiar with Microservices, event-driven architecture, and distributed systems, with practical experience applying these concepts in academic projects.

EDUCATION

Van Hien University

B.Sc. in Computer Science – Software Engineering | GPA: 3.2/4

Expected 2026

Ho Chi Minh City, Vietnam

- **Capstone Project:** Research and Development of an AI-Generated Image Forgery Detection System using Deep Learning and Explainable Artificial Intelligence (XAI)
- Clubs: Competitive Programming Club – OLP/ICPC Training Team

EXPERIENCE

Software Tester

HiAI Company

Feb 2026 – May 2026

Ho Chi Minh City, Vietnam

- Performed manual testing for web applications to identify functional defects and usability issues.
- Designed and executed test cases based on software requirements and user workflows.
- Tested RESTful APIs using Postman to validate request/response data and backend functionality.
- Reported, tracked, and verified bugs using Jira with detailed reproduction steps and expected results.
- Collaborated with developers to reproduce issues, retest fixes, and improve software quality before release.

PROJECTS

Learning Management System | GitHub

2026

- **Tech Stack:** Java 21, Spring Boot 4, Spring Security, Spring Data JPA, MySQL, Redis, JWT, OAuth2
- **Payment Module:** Built a multi-gateway payment module (VNPay, MoMo, ZaloPay) using the Strategy Pattern, so new payment providers can be added without touching existing code.
- **Payment Security:** Verified transaction integrity from asynchronous IPN webhooks using HMAC signatures (SHA-256/SHA-512) to detect tampered requests.
- **Authentication:** Combined JWT-based login with Spring Boot OAuth2 (Google Login) and role-based access control for protected endpoints.
- **Caching:** Added Redis caching for the course catalog and search queries to reduce database load under concurrent traffic.

AirGo – Flight Booking System | GitHub

2026

- **Tech Stack:** Java 21, Spring Boot 4, Spring Cloud (Gateway, OpenFeign), Kafka, Redis, MySQL, Docker
- **Microservices:** Split the backend into Booking, Seat, and Payment services, each with its own database, communicating through Kafka events.
- **Seat Reservation:** Used Redis TTL-based locks together with JPA optimistic locking (@Version) to prevent double-booking when multiple users reserve the same seat.
- **Data Consistency:** Implemented a Kafka-based Saga pattern to synchronize Booking and Payment services, combined with synchronous OpenFeign calls and scheduled rollbacks for Seat management.
- **API Gateway:** Added JWT authentication and Redis-based rate limiting at the API Gateway.

ACTIVITIES

ICPC Competitive Programming Team

Van Hien University

2024 – Present

Ho Chi Minh City, Vietnam

- Participated in algorithmic problem-solving training and competitive programming contests.
- Practiced data structures, algorithms, and problem-solving techniques for ICPC/OLP competitions.

SKILLS

Programming Languages: Java, C++, Python, JavaScript, SQL

Languages: Vietnamese (Native), English (Working proficiency)

Frontend: HTML, CSS, Bootstrap, Thymeleaf

Backend: Spring Boot, Spring MVC, Spring Security, REST APIs, JWT, OAuth2, JPA/Hibernate

Databases: MySQL, MongoDB, Redis

Tools & CI/CD: Git, GitHub, Docker, Postman, Swagger, GitHub Actions

PARTICIPANT

Olympic Tin Hoc Sinh Vien Viet Nam

2025

ICPC Asia Regional Contest | Ho Chi Minh City Site

2025